

**ECON 101—Introduction to Microeconomics
Spring 2023**

Midterm II

April 13, 2023

Name:

NU ID number:

Instructions:

- Answer all questions. Multiple choice answers go in the space provided below. You will be provided with scratch paper.
- The exam has a duration of 1 hour and 15 minutes, from 3 PM to 4:15 PM.
- You are not allowed to go out during the duration of the exam. This means no bathroom breaks. Thus, be prepared.
- You are only allowed to have in your desk area: ID, writing instruments, ruler, and a simple calculator (no graphing calculators), and a simple watch if needed (no smartwatches).
- You must leave all other personal items, including smart devices, to the side of the room. It may be a better idea to not bring them at all.
- If you finish early the exam, you will only be allowed to go out until 3:50 PM. Otherwise, you will have to remain in the room quietly until the exam is finished.
- You are not allowed to remove the staple from your exam booklet.

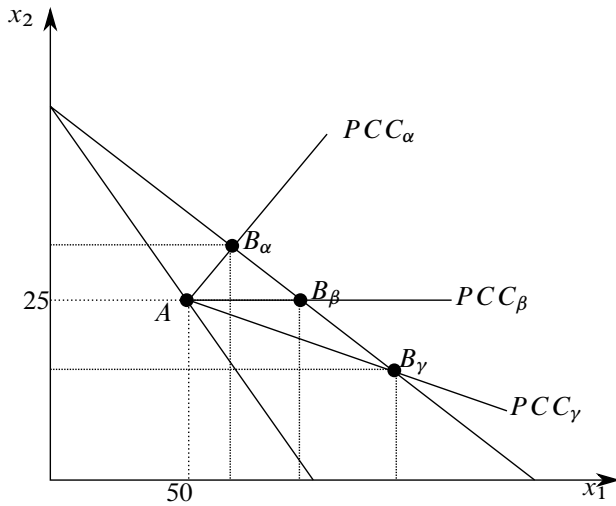
Multiple choice. Please write your answers here!

1	2	3	4	5	6	7	8	9	10

Do not write below this area!

Problem 1

Consider the following scenario as depicted by the figure below.



There are three consumers, denoted by α, β, γ which can consume two goods, x_1 and x_2 . The consumers

have the same Income of \$100 dollars. The price of x_1 is \$1 dollar and the price of x_2 is \$2. Each consumer initially chooses the initial consumption bundle $x_1 = 50, x_2 = 25$ which is denoted by point A. Suppose that there is a reduction in the price of x_1 to \$0.5. After this price decrease, the optimal consumption bundles are B_α, B_β , and B_γ , where the subindex denotes the respective consumer. In the graph you can see the depiction of the *price consumption curves* for each consumer.

Suppose that a market researcher tells you that after estimating the own price elasticities for the demand of good x_1 for each consumer, he has found the following:

- $|\epsilon_{1,p_1}| = 0.6$
- $|\epsilon_{1,p_1}| = 1.5$
- $|\epsilon_{1,p_1}| = 1$

Unfortunately, the market researcher has been very careless and he forgot to record which elasticity corresponds to each consumer.

1. For consumer α the goods are _____; for consumer β the goods are _____; and for consumer γ the goods are _____.
 - (a) neutral; complements; substitutes
 - (b) substitutes; complements; neutral
 - (c) substitutes; neutral; complements
 - (d) complements; neutral; substitutes
2. Now consider the cross-price elasticity for good 2, ϵ_{2,p_1} . For consumer α the cross-price elasticity is $\epsilon_{2,p_1} = \text{_____}$; for consumer β the cross-price elasticity is $\epsilon_{2,p_1} = \text{_____}$; and for consumer γ the cross-price elasticity is $\epsilon_{2,p_1} = \text{_____}$.
 - (a) -0.4 ; 0 ; 0.5
 - (b) -1.6 ; -2 ; -2.5
 - (c) 0 ; 0.5 ; -0.4
 - (d) -1.8 ; -1 ; 0
3. Consider now the budget shares $s_i = \frac{p_i x_i}{I}$ for each good $i = 1, 2$. At bundle A, the budget shares for the goods are _____ for all the consumers; at each bundle B for each consumer, the budget shares are: _____ for consumer α ; _____ for consumer β ; and _____ for consumer γ .
 - (a) different; $s_1 = s_2$; $s_1 = s_2$; $s_1 = s_2$
 - (b) equal; $s_1 < s_2$; $s_1 = s_2$; $s_1 > s_2$
 - (c) equal; $s_1 > s_2$; $s_1 = s_2$; $s_1 < s_2$
 - (d) different; $s_1 < s_2$; $s_1 = s_2$; $s_1 > s_2$

Problem 2

After a long week of telling his students to not succumb to panic (because of midterms) and many hours of battling against the university's administration, Prof. A (A is for *Anonymous*) likes to go to his favorite restaurant and relax by drinking one of his preferred cocktails: a Martini (M) or a Negroni (N). He considers the two cocktails as *perfect substitutes* but he gives an edge to the Negroni; specifically, his utility function is given by: $u(M, N) = M + 2N$. The price of a Martini is 3,000 tenge and the price of a Negroni is 4,000 tenge. Suppose that Prof. A's budget for going out for dinner is 12,000 tenge.

4. What will be Prof. A's choice?
- (a) He will drink 3 Negronis.
 - (b) He is equally fine drinking either 2 martinis and 1.5 Negronis OR only 3 Negronis.
 - (c) He will drink 4 Martinis.
 - (d) None of the above because Prof. A should be drinking soda instead.
5. After some consideration, Prof. A decides that he should do things responsibly: he should not drink more than 2 cocktails. That is, in addition to his budget constraint, he now has the extra constraint $M + N \leq 2$. What will be Prof. A's choice now?
- (a) He is equally fine drinking either 2 Martinis OR 1 Martini and 1 Negroni.
 - (b) He will drink 2 Negronis.
 - (c) He will drink 2 Martinis.
 - (d) None of the above because Prof. A should be having juice instead.

After thinking about it, Prof. A decides that he should like to prepare his Negronis at home. The Negroni recipe uses *three ingredients in equal proportions*: 1 part of the Italian liqueur Campari (C), 1 part of sweet Vermouth (V), and 1 part of Gin (G); serve in a chilled glass over ice and add an orange twist and you have a very nice cocktail. Considering only the three main ingredients, Prof. A's utility function for a Negroni can be written as $u(C, V, G) = \min\{C, V, G\}$. The price of a 1 liter bottle of Campari is 12,000 tenge, the price of a 1 liter bottle of sweet Vermouth is 6,000, and the price of a 1 liter bottle of Gin is 14,000 tenge.

6. If a standard cocktail is approximately 90 milliliters, what would be the cost (*the minimal level of expenditure*) of *one* home-made Negroni for Prof. A?
- (a) 960 Tenge.
 - (b) 1,280 Tenge.
 - (c) 3,200 Tenge.
 - (d) 0 because Prof. A should be drinking coffee instead.
7. How much more expensive is having a Negroni at the restaurant vs preparing it at home?
- (a) 316.67% more expensive.
 - (b) 25% more expensive.
 - (c) 212.5% more expensive.
 - (d) I don't know, but at least I learned that going out is very overpriced!

Problem 3

Consider the following scenario. You have two midterm exams coming up, and you have to decide how to allocate your time. After eating, sleeping, exercising, and maintaining some semblance of a social life, you have 10 hours each day in which to study for your exams, that is, 10 hours is your *total budget of study time*. You have figured out that your GPA from your two courses, Introduction to Microeconomics and Biology (for non-Science majors) takes the following form: $GPA = \frac{4}{7}(2\sqrt{E} + \sqrt{B})$, where E is the number of hours you devote to Economics and B is the number of hours you devote to biology. Both E and B should be regarded, for simplicity, as continuous variables. The marginal GPA with respect to Economics is $MGPA_E = \frac{4}{7\sqrt{E}}$, and the marginal GPA with respect to Biology is $MGPA_B = \frac{2}{7\sqrt{B}}$.

8. Compute the optimal amount of hours (E^*, B^*) that you should study for Economics and Biology.
 - (a) $(E^*, B^*) = (8, 2)$
 - (b) $(E^*, B^*) = (5, 5)$
 - (c) $(E^*, B^*) = (2, 8)$
 - (d) I don't know so probably I should have devoted the 10 hours to study Economics.
9. Now suppose that since Introduction to Microeconomics is a very hard class and it makes your head hurt, you realize that for every hour that you study of it, you need to take a one hour break. That is, if you study for E hours Economics, the actual time spent is $2E$. What is your optimal allocation of study time in this case?
 - (a) $(E^*, B^*) = (4, 2)$
 - (b) $(E^*, B^*) = (10/3, 10/3)$
 - (c) $(E^*, B^*) = (2, 4)$
 - (d) $(E^*, B^*) = (5/2, 5)$
10. Compute the Substitution Effect, the Time (*Income*) Effect, and the Total Effect of the increase in the opportunity cost (*price*) of studying Economics.
 - (a) The Substitution Effect is $20/9$, the Time Effect is $22/9$, and the Total Effect is $14/3$.
 - (b) The Substitution Effect is $-22/9$, the Time Effect is $-20/9$, and the Total Effect is $-14/3$.
 - (c) The Substitution Effect is $-20/9$, the Time Effect is $-22/9$, and the Total Effect is $-14/3$.
 - (d) The Substitution Effect is $22/9$, the Time Effect is $20/9$, and the Total Effect is $14/3$.
 - (e) I don't know, so definitively I should have devoted the 10 hours to study Economics.

Hint: Feel free to work with two decimals for this whole problem.